

SUMMARY

Data Engineer with 4+ years of experience in Python, SQL, and Bash, specializing in cloud-native ETL/ELT pipelines with Databricks, Kafka, Delta Lake, dbt, and Snowflake to build scalable, compliance-grade systems supporting equity-focused, trauma-informed work in mission-driven organizations. Committed to FERPA and HIPAA-aligned governance that strengthens community trust. Physicist and philosopher by training, I bring an ethics-informed lens to every system I build.

SKILLS

Programming Languages: Python, SQL, Bash, R, JavaScript/TypeScript, YAML, Jinja2, VBA, MATLAB

Analysis & Visualization: Tableau, Power BI, Excel, ArcGIS, Streamlit

Machine Learning: Scikit-learn, PyTorch, LangChain, OpenAI API, Vector Databases (Chroma), RAG, Prompt Engineering

Data Engineering: PySpark, Databricks, Delta Lake, dbt, Kafka, Snowflake, PostgreSQL, MongoDB, Hadoop, Apache Airflow

Cloud & DevOps: GCP (GCS, Dataproc, IAM), AWS (S3, EMR), Azure, Docker, Terraform, GitHub, GitHub Actions CI/CD, Bitbucket

PROFESSIONAL EXPERIENCE

DATA ANALYTICS & ENGINEERING INTERN

Partnership With Children

Jan 2025 – Jan 2026

- Engineered a **FERPA**-compliant automated ETL pipeline in **Python**, **JavaScript**, and **Bash** to migrate protected student records from Apricot into SharePoint, reducing a month-long manual workflow to 7 minutes while maintaining data integrity and real-time access
- Developed a secure **Azure**-hosted **Streamlit** dashboard with **Docker** and **Microsoft SSO** that provided leadership with reliable visibility into program performance, informing grant renewals, staffing allocation, and program expansion decisions across NYC schools, supported by **VBA** automation and **GitHub Actions CI/CD**
- Consolidated 10+ years of NYSED, InfoHub, and internal program datasets using advanced **SQL** joins and unions, validating and reconciling discrepancies to within 0.5% margin of error and standardizing metrics that directly supported grant reporting and district evaluations measuring social work impact across 30,000+ students in underserved NYC communities
- Migrated the entire Apricot case-management platform, including 10+ years of FERPA and **HIPAA**-sensitive student records, into Salesforce across 32 NYC schools, partnered with non-technical stakeholders to design **SOQL**-driven notification Flows that fit their actual workflows, and established data governance standards adopted organization-wide

NEUROAI DEEP LEARNING RESEARCHER

Neuromatch

May 2024 – Sep 2026

- Built research data pipelines across three Neuromatch cohorts processing **1.5M+ COCO-MS** image captions via image ID joins and one-hot encoding to feed **RoBERTa** and **BERT** models (2024), structured large-scale neural recording data for **GLM**-based behavioral decoding (2025), and implemented a dual forward-pass **Node Perturbation** pipeline benchmarked against **Backprop** and **Hebbian** baselines on **MNIST** (2026), supporting peer-reviewed publication ([arXiv:2409.09560](https://arxiv.org/abs/2409.09560))

DATA ENGINEERING APPRENTICE

Annalect

Aug 2022 – Nov 2022

- Optimized large-scale **Airflow**-orchestrated pipelines for pharma and media clients by restructuring **Jinja2** templates, modifying existing DAGs, and refactoring event-driven **AWS Lambda** trigger logic, reducing data latency by 30%
- Contributed to ETL/ELT workflow automation across AWS (Lambda, **EMR**, **S3**, **Redshift**, **Glue**) and **PostgreSQL**, leveraging **PySpark** and **Redshift Spectrum** to process terabyte-scale data, supporting pipelines driving a 10% CPA drop and 12% performance lift across client pipelines

DATA SCIENCE FELLOW

The Knowledge House

Sep 2021 – Sep 2022

- Built ETL pipelines in Python and Bash to ingest and clean 500+ alumni records, integrated **Google API** for user authentication and data ingestion, and conducted exploratory data analysis, including geospatial engagement patterns using Pandas, supporting a 20% increase in platform user engagement in a cross-functional team environment

PROJECTS

Real-Time Equity Analytics Platform | Kafka, Databricks, Snowflake, dbt

Oct 2025 – Dec 2025

- Designed and deployed a production-grade batch and streaming ELT medallion pipeline ingesting NYC School Climate and CDC Social Vulnerability Index data via Kafka and Dataproc into GCS Bronze/Silver layers, with Databricks Spark Structured Streaming and dbt-modeled Snowflake Gold marts powering district equity dashboards for policy and resource decisions
- Built a LangChain and Chroma RAG service with a FastAPI backend and React/TypeScript UI, embedding 5K+ SVI and school climate documents to enable natural language querying of equity metrics over validated Snowflake Gold data
- Provisioned the entire cloud infrastructure from scratch using Terraform across GCP, Snowflake, and Databricks, implementing DevSecOps practices including detect-secrets scanning, least-privilege IAM roles, and segregated GitHub Actions CI/CD with zero credentials in version control

Automated Compliance-Grade Document ETL Pipeline | Python, JavaScript, Bash

Mar 2025 – May 2025

- Built a FERPA and HIPAA-compliant ETL pipeline in Python, JavaScript, and Bash to extract, validate, and migrate 10,000+ protected student records from Apricot into Excel and SharePoint using Puppeteer-clustered PDF automation and a Tkinter GUI, reducing manual processing from months to minutes

PUBLICATIONS

Krotov, A., Tebo, A., Picart, D., Algave, A. (2024). *Evaluating authenticity and quality of image captions via sentiment and semantic analyses.* [arXiv:2409.09560](https://arxiv.org/abs/2409.09560)

Picart, D. (2018). *Method of Gas Detection Insensitive to Vibration.* [Wright's Research Lab](#)

EDUCATION

ADELPHI UNIVERSITY

Bachelor of Science, Physics; Minor, Philosophy

2014 – 2018

- Specialization in Engineering, AI, and Data Analysis with advanced mathematics foundations
- Year-Round NCAA D-II Cross Country and Track & Field Athlete

Engineering Fellowships & Continuing Experience

2022 – Present

- Program Earth Software Engineering Fellow (Sep 2024 – Present) · TKH Data Science AEP Fellow, Data Engineering Track (Sep 2024 – Dec 2024) · TEC Innovate Google Fellow (Oct 2024) · CUNY GIS Research (Oct 2023 – Dec 2023)